Online Social Networks and Media 15 (2020) 100058

Contents lists available at ScienceDirect

Online Social Networks and Media

journal homepage: www.elsevier.com/locate/osnem

Political communication on social media: A tale of hyperactive users and bias in recommender systems

Orestis Papakyriakopoulos*, Juan Carlos Medina Serrano, Simon Hegelich

Bavarian School of Public Policy, Technical University of Munich, Germany

ARTICLE INFO

Article history:
Received 20 May 2019
Revised 10 October 2019
Accepted 22 October 2019
Available online 25 December 2019

Keywords:
Social media
Political communication
Hyperactive users
Recommendation systems
Algorithmic bias
Recommender systems
Topic modeling
Computational social science
Political data science
Agenda setting

ABSTRACT

A segment of the political discussions on Online Social Networks (OSNs) is shaped by hyperactive users. These are users that are over-proportionally active in relation to the mean. By applying a geometric topic modeling algorithm (GTM) on German users' political comments and parties' posts and by analyzing commenting and liking activities, we quantitatively demonstrate that hyperactive users have a significant role in the political discourse: They become opinion leaders, as well as having an agenda-setting effect, thus creating an alternate picture of public opinion. We also show that hyperactive users strongly influence specific types of recommender systems. By training collaborative filtering and deep learning recommendation algorithms on simulated political networks, we illustrate that models provide different suggestions to users, when accounting for or ignoring hyperactive behavior both in the input dataset and in the methodology applied. We attack the trained models with adversarial examples by strategically placing hyperactive users in the network and manipulating the recommender systems' results. Given that recommender systems are used by all major social networks, that they come with a social influence bias, and given that OSNs are not per se designed to foster political discussions, we discuss the implications for the political discourse and the danger of algorithmic manipulation of political communication.

1. Introduction

To day, the internet prevails as a prominent communication and information medium for citizens. Instead of watching TV or reading newspapers, increasing numbers of people become politically informed through online websites, blogs, and social media services. The latest statistics demonstrate that internet as a news source has become as important as television, with its share increasing year by year [1]. Given this shift in the means of news broadcasting, politicians have altered their tactics of communication to the society. OSNs, such as Twitter, Facebook and Instagram, have become a cornerstone of their public profiles as they use OSNs to transmit their activities and opinions on important social issues [2–4].

The growth of online communities on social media platforms have created a public amenable to political campaigning. Political parties and actors have adapted to the new digital environment [5], and besides new campaigning methods such as microtargeting [6], have created microblogs through which they can inform citizens of their views and activities. Furthermore, OSNs have enabled users to respond to or comment on politicians' messages, giving birth to a new type of political interaction and transforming the very nature of political communication itself.

On OSNs, the flow of information from politicians to citizens and back follows a different broadcasting model than the classical one [7]. Instead of journalists monitoring the political activity, political actors themselves produce messages and make them publicly available on the platforms. Each platform provides its users with access to the generated content, as well as distributes it to them through recommendation algorithms [8,9]. The received information is then evaluated both directly or indirectly [10,11]: The political message is interpreted immediately, or subsequently through further social interactions among citizens on the related topics. On OSNs, not only can users respond to politicians in the traditional manner i.e., through their political activity in the society-, but also respond to or comment on the politicians' views online. This creates a new type of interactivity, as users, who actively engage in online discussions sharing their views, are able to influence the way the initial political information will be assimilated by passive users as well as directly influencing political actors.

This new form of political interactivity transforms political communication. Given the possibility of users to directly respond

* Corresponding author.
E-mail addresses: orestis.papakyriakopoulos@tum.de (O. Papakyriakopoulos), juan.medina@tum.de (J.C.M. Serrano), simon.hegelich@hfp.tum.de (S. Hegelich).

https://doi.org/10.1016/j.osnem.2019.100058

to the political content set by political actors, and discuss political issues with other citizens online, OSNs emerge as a fruitful space for agonistic pluralism. They provide the necessary channels for different interests and opinions to be expressed, heard and counterposed; elements that constitute the very essence of political communication. If the discussions held are legitimized within a democratic framework, they form the basis for reaching a conflictual consensus [12], based on which societal decisions can be made. Hence, political communication on OSNs opens new possibilities for citizens to participate in the political shaping of the society, providing them with additional space to address their interests.

1.1. Motivation

Although the above type of political communication has the potential to improve the function of democracy, OSNs possess a structural property that obstructs the unbiased constructive interaction between political actors and citizens: The activities of users on OSNs follow an extreme value distribution [13–16]. Practically, this means that users are not equally active when using a specific OSN. Among others, the majority of users remain passive, or participate with a very low frequency; they either simply read the content or like, comment, tweet, etc. very rarely. On the contrary, a very small number of the users are hyperactive, as they over-proportionally interact with the platform they use. Thus, in political communication on OSNs, hyperactive users are citizens who over-proportionally externalize their political attitudes compared to the mean. This could be done by liking, commenting, tweeting or using any other interaction possibility provided by a platform to declare an attitude to a political issue.

The given activity asymmetry becomes a major issue, considering that a considerable part of the society is politically informed via OSNs. As hyperactive users externalize their political attitudes more than the others, they have the potential to distort political communication; political issues that are important to them become overrepresented on OSNs, while the views of normally active users become less visible. Hence, hyperactive users may influence the political discussions towards their ends, creating a deformed picture of the actual public opinion on OSNs. This fact violates the assumption of an equitable public political discourse as part of political communication [17], because the interests and views of normally active users appear as less important.

The above distortion of political communication is intensified by the business models of the OSN platforms. OSNs were not created to be arenas of political exchange. Their aim is to maximize the number of platform users, by keeping them satisfied [18], and to transform this social engagement to profits, i.a. through advertisement. Hence, on OSNs, users are both consumers and citizens [19]. In order to maximize their profits, OSN platforms adjust their recommendation algorithms to the content popularity, with a view to promoting information that most users will like. For example, the recommendation algorithm of Facebook aims to maximize how well users spend their time on the platform [20]. This is translated to the promotion of content that influences users to engage more with the website, i.e to generate more likes, comments or shares. Facebook also gives further incentives for users to engage more on the platform. It rewards users, giving them badges as 'top commenter' or 'top fan' that appear next to their name. This happens when they are really active on the platform or on a specific page. These awards give users a higher social status on the platform and actually aim the mobilization of users to engage more with the service. This can be then translated to more user data for the platform and consequently, yields better placed advertisement.

As hyperactive users asymmetrically influence the popularity of political content, recommender algorithms might also replicate this asymmetry. A platform might recommend content which is consistent with the political interests of hyperactive users. This phenomenon per se denotes a form of algorithmic manipulation of political communication: The platform unintentionally magnifies hyperactive users' interests, thus posing the risk of political invisibility for passive users [21]. The investigation of such an issue becomes politically important, considering that the most engaged news stories on social media platforms usually come from right-wing media sources and contain provocative content [5,20]. Nevertheless, understanding the interplay between hyperactive users and recommendation systems is not a trivial task. OSNs conceal exactly how their models work, as well as their effects on user behavior. Given these restrictions, the study tries to shed light on potential algorithmic distortions of content, partly by performing simulations that try to replicate real world OSN features.

Last but not least, aforementioned misrepresentations of public opinion have a direct impact on political campaigning. Contemporary political actors develop their influence strategies based on the perceived voter model [22], which presupposes the gathering of demographic and political data for the development of statistical models about the electorate's attitudes. As major part of these data is derived from social media, models that fail to take the effect of hyperactive users into account would face an important bias.

1.2. Research questions

Considering the above, we want to answer following questions:

RQ1: What is the impact of hyperactive users on political agenda-setting on OSNs?
RQ2: Does hyperactive behavior influence recommendation algorithms?

To coherently provide answers to the above questions, we analyze a dataset of posts and user reactions generated on political pages on Facebook. We classify the generated content into topics by applying a topic modeling algorithm and formulate five hypotheses to be tested.

For the investigation of the impact of hyperactive users on political agenda-setting, we want to understand: A. How the user activities are distributed on the political pages. B. If hyperactive users are interested in the same content as regular users. C. If hyperactive users' opinion is equally, more, or less important in the discussions taking place. Therefore we test the following hypotheses:

H1: The log-normal distribution is the optimal distribution for mathematically describing the users' liking and commenting activities on political pages on Facebook.
H2: The distribution of topics in which hyperactive users engage does not deviate from the distribution of topics in which the regular users engage.
H3: Hyperactive users' comments receive on average the same amount of likes as the comments of the rest of the users.

Besides understanding the role of hyperactive users on political discussions, we want to quantify the contribution of hyperactive users on the suggestions made by recommendation systems. We focus on two parts of the recommendation systems. A. Their training process, and how efficient different models are in learning from heavy-tailed distributions of political activities. B. Given that algorithmic recommendations on OSNs depend on the users' friendship network, we investigate how the suggested content can be manipulated by poisoning the friendship graphs, i.e. by adding users that show hyperactive behaviour. We achieve this by testing following hypotheses on a simulated political network:

H4: Given the heavy tailed distribution of activities and data sparsity, standard recommendation algorithms fail to suggest content that correspond to the users' political interests.

H5: Given the interconnectedness of recommender systems and friendship network structure, graph poisoning by the introduction of hyperactive users significantly distorts recommendations of political content.

1.3. Original contribution

- We mathematically define hyperactive users on OSN Facebook, and identify them on the public pages of the major German political parties.
- By applying a state-of-the-art topic modeling algorithm, we investigate whether they spread or like different messages on political issues other than normal users and politicians do. We prove that hyperactive users are not only responsible for a major part of online political discussions, but also externalize different attitudes than the average user, changing the discourse taking place.
- We quantify their effect on content formation by measuring their popularity and showing that they adopt an opinion leader status.
- We simulate a social network of political activities and train hybrid collaborative filtering and deep learning recommendation algorithms on it. We prove that recommendations are strongly influenced by hyperactive behavior. We also show that the election of an appropriate cost function can deal with the issue.
- We insert adversarial examples of hyperactive users and intentionally manipulate the recommender systems' suggestions. Given the above, we initiate an important discussion on OSNs as spaces of political communication.

2. Background and related work

Research proves that user activities on OSNs are not normally distributed. The interactivity time of users with the services, the frequency with which they generate content on the platforms, how often they like, tweet or comment, all follow an extreme value distribution [13,14,23]. This denotes a significant asymmetry existing between users' behaviour, which stops being a mere descriptive behavioral feature when it comes to analyzing phenomena as political communication. The fact that users contribute differently on political discussions and on news diffusion automatically becomes a political phenomenon, which needs to be analyzed by researchers. This is crucial for understanding how political processes take place in the digital era.

The given asymmetry becomes even more important when considering that most OSN users remain passive. They just browse and read political content [16], without contributing in information diffusion [24]. This automatically constitute active and hyperactive users as potential opinion influencers, as they shape the platforms' content. Research proves that users who obtain influential roles in networks have the ability to politically persuade the rest [25] as well as to guide information diffusion [26] in terms of news stories and political discussions [27]. Therefore, it is interesting to investigate and define the differences of normally active and hyperactive users, and their role in shaping political content. Hyperactive behaviour constitutes a widely used tactic in information operations, with automated or human accounts intensively trying to influence discussions taking place [28,29].

The extremely skewed distribution of user activities also becomes an issue when considering the function and impact of recommendation systems. All major OSNs deploy data-intensive algorithms for suggesting to the users contents to interact with, especially on the platforms' news feed. These algorithms consist usually of deep neural systems that take as input user specific features and return contents that maximize the probability of a user interacting with it [30]. Although most OSNs do not disclose the exact structure of their recommender systems, it is known that they use the users' social network structure to improve recommendations' accuracy [31,32]. In this way they take advantage of the homophily principle that states that a user's friends will have similar interests as the user themselves.

Nevertheless, research proves that the training of statistical models and neural architectures on extremely skewed and imbalanced datasets as that of user activities might yield poor accuracy results [33,34]. Therefore, the danger exists that algorithms might fail to learn and diffuse in a coherent way the users' political interests, resulting to the algorithmic manipulation of political communication [21]. To that phenomenon might contribute the fact that recommender systems come with a social influence bias: They actually have the power to alter public opinion [35,36].

Besides having the potential to influence the recommender systems' training process, hyperactive behavior is able to distort the recommendations of already trained models. Researchers prove that adversarial attacks of neural architectures used for recommendation systems can make the models fail. This can be done by both inserting adversarial observations at the level of the model, or by graph poisoning: altering the network structure in a way that specific information will become invisible or be overrepresented [36–40]. Given the number of hyperactive users that explicitly aim the diffusion of political content, it is important to investigate their impact on recommender systems' results.

3. Data and methods

The section provides an overview of the data and methods used to study the impact of hyperactive users on political agenda-setting (RQ1) and to test if hyperactive behavior influences recommendation systems (RQ2). Subsection 3.1 provides an overview of the data used to test the five formulated hypotheses and explains the related limitations. Subsections 3.2-3.4 present the tools used to quantify the presence and activities of hyperactive users on Facebook political pages (RQ1). For answering RQ2, subsections 3.4 and 3.5 provide details on the development of recommendation systems based on user activities and their differences and similarities from the actual algorithms used by OSNs. We describe how the systems make recommendations and how we attack the systems with adversarial examples to potentially distort their recommendations.

3.1. Data description & limitations

To investigate the effect of hyperactive users, we analyse the public Facebook pages of the main German political parties. Our sample includes CDU, CSU, SPD, FDP, Bündnis 90/Die Grünen, Die Linke, and AfD. CDU is the main conservative party of Germany, while CSU is the conservative party active in Bavaria. SPD represents the main German social-democratic party, and Die Linke the radical left. AfD has a nationalist, anti-immigrant, and neo-liberal agenda, while FDP is a conservative, neo-liberal party. Finally, Bündnis 90/Die Grünen is the German green party. We focus on Facebook, because the platform's API restrictions and its monitoring system largely prevent automated activities, such as those performed by social bots on other platforms [41,42]. Therefore we can evaluate the natural behavior of hyperactive users and not an algorithmically generated one.

We take into consideration all party posts and their reactions in the year 2016. This choice is made, because the acquirement of similar data is not feasible today, since Facebook has restricted access to its public API. In total, by accessing the Facebook Graph API in 2017, we collected 3,261 Posts, 3,084,464 likes and 382,768 comments, made by 1,435,826 users. The sample includes all posts

Table 1
Overview of activities on the 7 investigated Facebook pages for the year 2016. Ratio denotes the total number of likes and comments made on each page to the total number of active users.

	Subscribers	Active users	Posts	Comments	Likes	Reactions to users ratio
AfD	309,275	642,927	577	176,506	1,601,502	1,778,008 : 642,927
CDU	123,534	77,400	391	46,837	82,564	129,401 : 77,400
CSU	148,759	404,641	595	80,618	692,273	772,891 : 404,641
Die Grünen	132,630	135,217	209	24,750	153,500	288,717 : 135,217
Die Linke	167,570	135,085	356	14,257	181,161	195,418 : 135,085
FDP	56,581	76,919	579	9,177	126,584	135,761 : 76,919
SPD	121,128	179,970	566	30,630	246,887	277,517 : 179,970

Table 2
Vuong test results.

Log-normal vs	Likes LL-ratio (p-value)	Comments LL-ratio (p-value)
Power-law	15.1 (<0.01)	34.9 (<0.01)
Poisson	34.9 (<0.01)	12.7 (<0.01)
Exponential	12.7 (<0.01)	26.6 (<0.01)

and comments on the posts generated for the period under investigation.

Table 1 provides an overview of the data collected. We only concentrate on 7 German political pages, and we do not investigate the numerous other smaller party pages existing on the platform. Nevertheless, because our sample includes activities of more than 1,4 million people, we can draw consistent inferences about the statistical distribution of users related to political communication. From Table 1 it is visible that activities among the pages are not evenly distributed. This conforms with existing literature stating that AfD users are the most active German partisans online [43]. It also agrees with existing literature stating that OSNs come with a population bias: user preferences on OSNs do not describe accurately preferences of people offline [44]. This asymmetry does not influence our investigation. Although some pages might be more popular than others, and user behaviour might vary in between, our investigation focuses on revealing statistical properties of user activities on the party pages as an integrated ecosystem. This fact also deals with the limitation that we do not have the activities of users on other non-political pages. Our analysis deals only with the interaction of users with political content and wants to uncover features and issues of political communication taking place on OSNs.

3.2. Defining hyperactive users

For testing if the log-normal distribution is the optimal distribution for mathematically describing the users' liking and commenting activities on the investigated pages (H1), we make theoretic assumptions about hyperactive users and choose the formal framework that best complies with their properties.

We consider hyperactive users as people, whose behavior deviates from the standard on an OSN platform. To obtain an understanding of the overall behavior of the users, we fit discrete power-law and extreme value distributions to mathematically describe the users' like and comment activities. Additionally, we run bootstrapped and comparative goodness-of-fit tests based on the Kolmogorov–Smirnov [45] and the Vuong [46] statistic to evaluate the potential fits, as proposed by Clauset et al. [47]. The KS test examines the null hypothesis that the empirical sample is drawn from the reference distribution, while the Vuong test measures the log-likelihood ratio of two distributions and, based on it, investigates whether both empirical distributions are equally far from a third unidentified theoretical one.

In order to mathematically describe the activities of hyperactive users, we select to treat them as outliers of the standard OSN population. We adopt the definitions made by Barnett and Lewis [48], Johnson and Wichern [49] and Ben-Gal [50], and see outliers not as errors, or coming from a different generative process, but as data containing important information, which is inconsistent with and deviating from the remainder of the data-set. Therefore, given the extreme skewed distribution of the activities, we follow the method proposed by Hubert and Vam der Veeken [51] and Hubert and Vandervieren [52] for outlier detection. We calculate the quartiles of our data Q_1 and Q_3, the interquartile range $IQR = Q_3 - Q_1$ and the whiskers w_1 and w_2, which extend from the Q_1 and Q_3 respectively to the following limits:

$$[Q_1 - 1.5e^{-4MC}IQR, Q_3 + 1.5e^{3MC}IQR] \tag{1}$$

where MC is the medcouple [53], a robust statistic of the distribution skewness. Data beyond the whiskers are marked as outliers.

3.3. Topic modeling

After evaluating the likes and comments distributions, as well as identifying the existing hyperactive users, we prepare our data for the application of a topic modeling algorithm. By applying a topic modeling algorithm and assigning user activities to the different topics, we can test if the distribution of topics in which hyperactive users engage does not deviate from the distribution of topics in which the rest of the users engage (H2).

As it has been shown that a noun-only topic modeling approach yields more coherent topic-bags [54], we clean our posts and comments from the remaining part-of-speech types. To do so, we apply the spaCy pretrained convolutional neural network classifier [55] based on the Tiger [56] and WikiNER [57] corpuses, classify each word in our document collection, and keep only the nouns.

We want to investigate the various topics that users and parties discussed but do not want to differentiate on the way they talked about them. Parties usually use a more formal language when posting on a topic than users. Therefore, there is the risk that the topic modeling algorithm would create different topics on the same issue, one for the parties and one for the users. To avoid this, we fit our model on the user comments, and then classify the parties' posts through the trained model.

For our analysis, we apply a non-parametric Conic Scan-and-Cover (CoSAC) algorithm for geometric topic modeling [58]. Our decision is based on the fact that most topic modeling algorithms (e.g. LDA [59], NMF [60]) need a priori as input the number of topics to split the corpus. CoSAC has the advantage of electing itself the number of topics to find the most efficient topic estimates.

The algorithm presupposes that the optimal number of topics K are embedded in a V-1 dimensional probability simplex Δ^{V-1}, where V the number of words in the corpus. Each topic β_K corresponds to a set of probabilities in the word simplex. The totality of topics build a convex polytope $B = conv(\beta_1, \ldots, \beta_K)$. Each document corresponds to a point $p_m = (p_{m1}, \ldots, p_{mV})$ inside Polytope B, with $p_m = \sum_k \beta_k \theta_{mk}$. θ_{mk} denotes the proportion that topic k covers in document m. Finally each document is drawn from a multi-

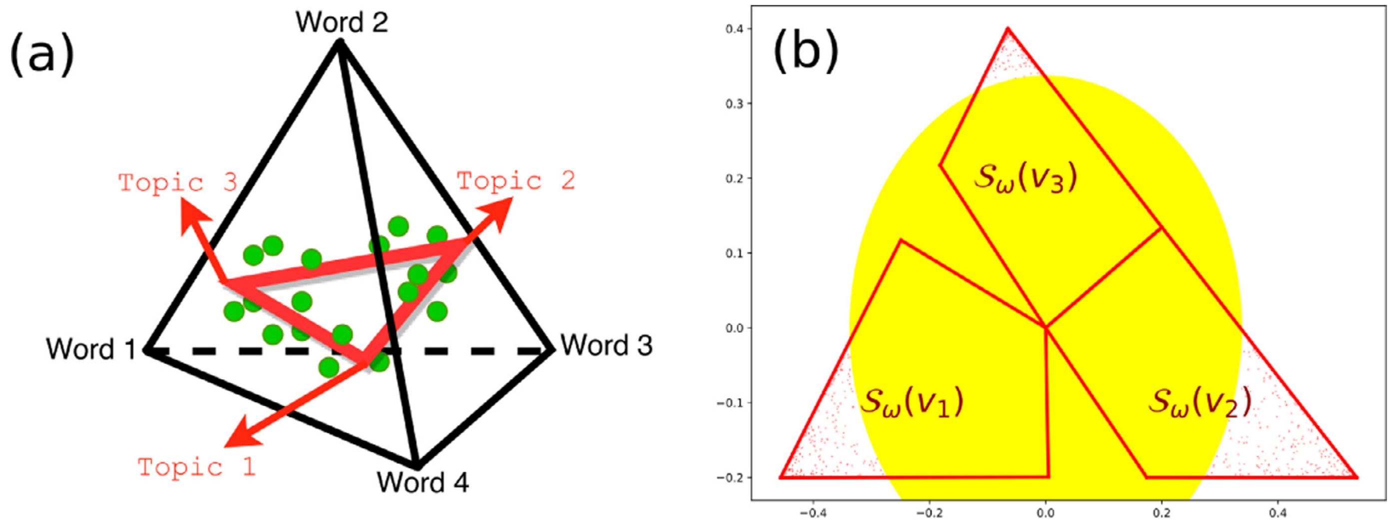

Fig. 1. (a) The topic polytope embedded in the word simplex. (b) Cones and sphere coverage of the polytope (figures from [58]).

nomial distribution of words: $w_m \sim Multinomial(p_m, N_m)$, where N_m the number of words in document m.

The CoSAC algorithm iteratively scans the polytope B and finds the furthest point from its center Cp. It then constructs a conical region with angle ω, starting from Cp and embedding the specific point (Fig. 1). All points within the cone are considered to belong in the same topic and are removed from the polytope. The procedure is iterated K−1 times, until almost no points remain in the polytope. A cone is considered sufficient if it covers at least a proportion of documents λ. After fitting the cones, CoSAC places a sphere with radius R to the polytope, to cover the remaining points. The K geometric objects and their respective points correspond to the K topics created by the algorithm. In our model, the hyperparameters were set to $\omega = 0.6, \lambda = 0.001$ and $R = 0.05,$ as proposed by the authors.

3.4. Comparison of activities

Given our topics, we want to evaluate the differences in the activity of normal and hyperactive users (H2). Therefore, we calculate the empirical distributions $f(comment|topic)$ over all topics for the comments of normal and hyperactive users respectively. We pairwise compare the distributions for each topic, by applying a 2-Sample Anderson–Darling Test [61]. The test calculates the probability that the populations from which two groups of data were drawn are identical.

Besides testing the empirical comment-topic distributions, we assign to each comment the topic with the highest probability and compare the most commented topics for normal and hyperactive users. Similarly, we assign the classified party posts to their most probable topic and aggregate the likes of normal and hyperactive users. In this way, we are in the position to locate the concrete political interests of users.

To complete the investigation of RQ1, we calculate the average likes that the comments of hyperactive and normally active user' comments received. In this way, we test H3, namely if hyperactive users' comments receive on average the same amount of likes as the comments of the rest of the users.

3.5. Training recommendation algorithms

In this and the next subsection we describe the methodology followed to answer our second research question: Does hyperactive behavior influence recommendation algorithms? Here, we describe two types of recommendation systems developed to test H4, namely if the heavy tailed distribution of user activities influences standard recommendation algorithms in learning the actual political interests of users.

We explain our methodology and describe differences and similarities between our recommendation systems and the actual ones' used by OSNs. Our algorithms do not fully correspond to the one's actually deployed by Facebook. We are not in the position to do that, because Facebook and most platforms do not disclose the exact systems they use. Nevertheless, by training recommendation algorithms that share common properties with the actual ones, we want to illustrate potential issues related to political communication on OSNs, as well as point out the need for additional transparency on the algorithms used by the platforms and their impact on user behavior.

To test the effect of hyperactive behavior on the suggestions of recommendation algorithms, we simulate a social network of political activities. We use the Facebook social circles dataset developed by [62], which contains a real network of 4,039 Facebook users and their corresponding 88,234 friendship relations. The dataset also provides anonymized user metadata about their age, sex, education, work, hometown and location. The network consists of 10 communities. We merge them into 7, in order to correspond to the number of party pages we investigate. For each community, we only map users active on a specific party page to the nodes, maintaining homophilic features initially existing in the network. Homophily states that similar users tend to form ties with each other [63]. We preserve this similarity by keeping the meta-data and connections from the original dataset and extend it by adding an additional common feature i.e. the political page users are active on.

In total, we create a social network that includes users' political attitudes and nonpolitical properties, while also having a friendship structure corresponding to an actual Facebook sub-network. We do not assign to a user all the posts they reacted on, but use the developed topics as proxies of the user's interests i.e., we assign how many times a user liked a post from a specific topic or commented on a specific topic. We built recommender systems by using the tensorrec library [64] based on the users' meta-data, the user's political interests and the political interests of their friends. In this way we tried to replicate as closely as possible the function of the private recommendation systems of OSN platforms, which take into consideration user features, location settings and the activities of friends in order to tailor news feed suggestions [31,65,66].

Of course, the actual OSNs are trained online and have the actual posts users have interacted with or might interact with in the future, as parameters. They also take as input thousands of features we do not have access to. For example, they take into consideration the pages a user has made a meaningful interaction, regardless of it is related to politics or not. They also include features about the general popularity of content generated on the pages, how much time a user spends on specific content, what information they share on other services of the platforms, or what device a user uses. All these pieces of information are not available to us, and instead of speculating about their impact we leave them out of our analysis. Nevertheless, we train recommendation algorithms on the available data and illustrate the impact of hyperactive behaviour. Training models on topics as proxies of the actual posts does not reduce the credibility of the study, as we want to make inferences about the impact of the models on agenda-setting and their accuracy on suggesting political content (H4,H5).

We train two models, a hybrid collaborative filtering model (HCF) and a deep learning recommendation system (DNN-BWMRB). The first one maintains properties of the recommender systems used on OSNs, i.e. it takes into consideration user features and friends activities, but is not specially designed for extremely skewed input data. By contrast, DNN-BWMRB uses a special cost function to deal with sparse and asymmetric input and output data.

The HCF optimizes the cost function

$$L = \min_{W,V} |P - UW * IV|_F$$

where P the n by t matrix containing the number of interactions of n users on the t different topics; and U the n by m matrix of user meta-data containing m features per user and I the t by n friends' preferences matrix containing for each user how many times their friends interacted about a specific topic. W and V are n by t and t by t parameter matrices to be optimized, $*$ the dot product, and $|.|_F$ the frobenius norm. Similarly, DNN-BWMRB consists of two parallel architectures with 3 fully connected layers of 276 neurons each, which take as inputs matrices U and I^T and minimize the balanced weighted margin-Rank Batch loss [67] (BWMRB) of predicting matrix P. BWMRB optimizes the rank-sensitive cost function

$$L_{BWMRB} = \min \sum_{j=1}^{n} \sum_{k=1}^{t} ln\left(\frac{n_{j,k}}{pop(k)} rank(j,k) + 1\right)$$

where j is a user, k a topic, $n_{j,k}$ the interaction of user j with content related to topic k, where $pop(k)$ is the total number of interactions on topic k. Also, $rank(j, k)$ is the ranked importance of a topic for a user given by the negative sampling equation

$$rank(j,k) = \frac{|K|}{|Z|} \sum_{k' \subset Z} \max(1 - p(j,k) + p(j,k'), 0)$$

where $p(j, k)$, $p(j, k')$ the scores predicted for user j, topic k, user j and a negative topic k' respectively. $|Z|$ is a set of randomly selected negative samples and $|K|$ the set of all topics. We select the specific cost function because it is robust against asymmetric interactions of unique users on the topics, as well as total popularity asymmetries between topics. An overview of the architecture can be found in Fig. 2.

3.6. Designing attacks with adversarial examples

The last part of our study focuses on understanding the vulnerability of recommender systems to attempts of organized manipulation. This concludes the investigation of how hyperactive behavior might influence recommendation systems (RQ2) by testing if graph poisoning by the introduction of hyperactive users significantly distorts recommendations of political content (H5).

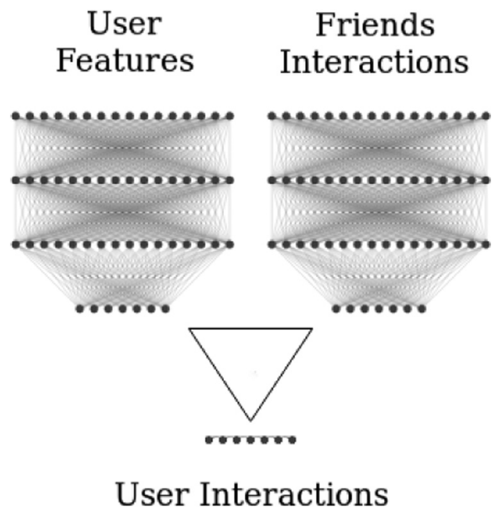

Fig. 2. Structure of DNN-BWMRB, consisting of two parallel neural networks leading to the balanced weighted margin-Rank Batch loss function.

We measure how models' recommendations change in the presence of adversarial examples that attempt to set the political agenda. After training the recommender systems on the initial network, we change the network structure by inserting new users, who have purposefully intensive interest on topic 10 i.e., on the deportation of immigrants. We perform a detailed sensitivity analysis, by varying the number of reactions they make on the topic, the number of users involved in the attack, as well as their position in the network. For that, we calculate the network eigenvalue centrality for each user in the friendship network, and adding the users in positions of high, moderate and low eigenvalue centrality. We sample from the already trained recommendation system's random suggestions, giving as input the initial and the attacked network. As recommendation algorithms take users' friends' interests as input, their suggestions adjust in respect to changes in the friendship network structure. By performing the aforementioned attack, we quantify via counterfactual scenarios how the suggestions made by the systems shift in the presence of adversarial examples.

4. Results

Our results are split in two parts, each one of them dedicated to answer a research question and its related hypotheses. The first part investigates hypotheses related to RQ1, i.e. on the impact of hyperactive users on agenda-setting on OSNs. In the second part we investigate if hyperactive behaviour impacts recommender systems' suggestions (RQ2).

4.1. What is the impact of hyperactive users on political agenda-setting on OSNs?

In order to address RQ1, we present first our findings on the general user distribution of the investigated pages and answer if the log-normal distribution is the optimal one to describe the data (H1). Based on that, we analyze the number and distribution of hyperactive users among the different pages. Then, we compare the behavior between hyperactive and normal users by taking into consideration the topic modeling results and further statistical tests. In this way, we both test whether hyperactive users engage on the same content with the rest (H2) and if they are equally popular to the other users (H3).

As a first result, we identify the log-normal distribution as the best measure for describing user activities (Fig. 3), confirming thus

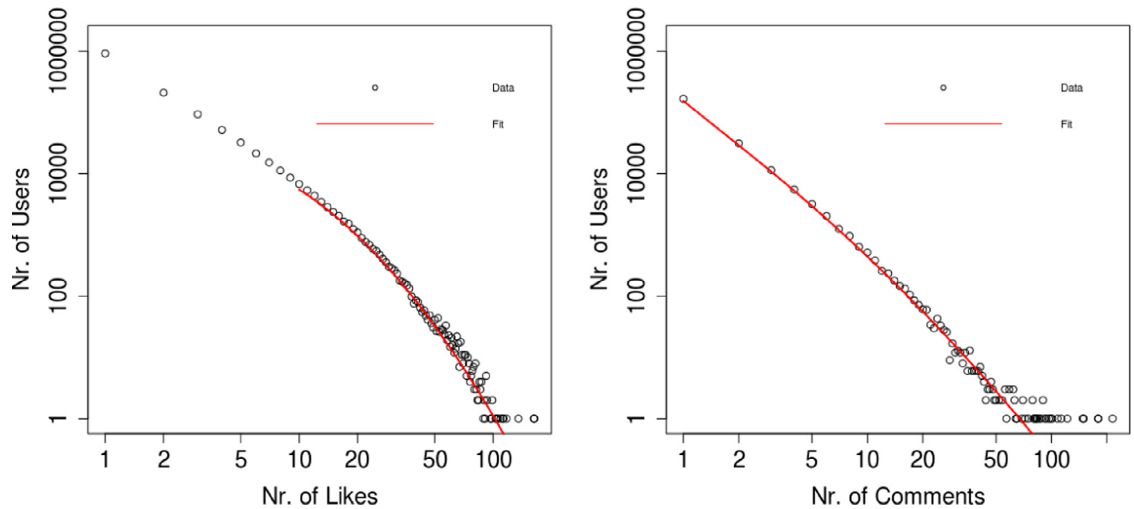

Fig. 3. Distribution of users' commenting and liking activites. The red line representes the log-normal fit.

H1. The bootstrapped KS-Tests (100 samples, two tailed) for both comments and likes fail to reject the null that our data come from a log-normal distribution (gof $<$ 0.01, p $>$ 0.05 and gof $<$ 0.01, p $>$ 0.2 respectively), while comparative Vuong tests show a better fit of the log-normal in comparison to the power-law, poisson and exponential distributions (Table I). Our results comply with the existing literature, which states that usually complex social network properties are log-normally distributed [15,68,69]. Fig. 3 shows the empirical frequencies of user activities and their respective log-normal fits.

After finding the distribution that best describes our data, we categorize users in hyperactive and normally active. Through our outlier detection methodology, we detect 12,295 hyperactive users on the comment section of pages, who correspond to 5.3% of the total users commenting on the pages. Due to the extreme skewness of the comments' distribution, a user is characterized as hyperactive if they made three or more comments. This is justified by the fact that 74% of the users under investigation made only one comment. Although hyperactive users represent 5.3% of the total commenting population, they account for 25.8% of the total comments generated on the parties' pages. Furthermore, 56% of these hyperactive users commented on two or more party pages, denoting that they generally interact with users across the political spectrum. By evaluating the popularity of the users' comments, we find that hyperactive users tend to get more support than the rest. Comments made by hyperactive users on average gained 3.52 likes, while normal users' comments only 3.07, a difference that is statistically significant (Mann-Whitney Test with continuity correction, one tailed: $W = 1.4^{10}$, $p < 0.01$). Thus, we reject the hypothesis that hyperactive users' comments are liked equally often as normal users' comments (H3). on the contrary, we show that hyperactive users are more appreciated in their contributions. This complies with previous research stating that highly active users tend to have the characteristics of opinion leaders [25].

Similarly, the evaluation of the pages' likes results in the characterization of 61,372 users as hyperactive, or 4.3% of the total users that liked the parties' posts. As before, the methodology labels users as hyperactive if they made three or more likes, because the majority of the active Facebook population rarely interacted with the related pages. The likes of these hyperactive users account for 26.4% of total likes, thus having a major effect on the overall content liked. In addition, 29% of hyperactive users like posts of more than one party, denoting again that their activities are spread over the entire parties' network. The overview of the hyperactive users' commenting and liking activities for each party can be found in Table 3. We also compare the like and comment distributions, by calculating their gini index. The measure provides a proxy for inequality, with 0 denoting perfect equality and 1 extreme inequality. In our case, we calculate a value of 0.35 and 0.45 for the comment and like distribution respectively. This denotes that like activities are more unequally distributed than the comment activities i.e., hyperactive users play a bigger role in the formation of likes. Additionally, the values denote a degree of inequality between normal and hyperactive users, but not an extreme one. Nevertheless this is misleading, because the measure does not take into consideration inactive users. Had that information been included, the gini index would have been much higher in both cases.

Based on the categorization of users as hyperactive or normal, we evaluate the results of the topic modeling algorithm. The model clustered the users' comments into 69 main topics. A major part of the topics concern the refugee crisis of 2016 and the related discussions about Islam. A set of topics aggregate comments on German Chancellor Merkel, on the leaders of other parties, on female and male politicians and the German parties in general. There is one topic summing up comments in English language, as well as a topic containing hyperlinks. Furthermore, the algorithm created policy related topics regarding foreign affairs, as well as the economy and labour market and the state in general. Other topics are related to the German national identity, society and solidarity, and the nature of democracy. Users also talk about family and gender policy, homeland security, transportation and environmental policy. There are topics that include wishes, fear, ironic and aggressive speech, as well as topics aggregating user thoughts. Finally, a set of topics is about political events and communications and a number of topics include comments against mainstream media and the political system.

The GTM algorithm is able to provide a broad picture of the discussion topics on the parties' pages, revealing numerous insights about the way Facebook users commented on the parties' posts. By splitting the comments into two categories, one for the ones generated by hyperactive users and one for the comments of normal users, and by assigning them to the topics to which they were mostly related, we create a a heatmap that provides a qualitative overview on the activities across pages (Fig. 4). The heatmap illustrates which topics are prominent on the users' comments on each political page. Users on the AfD page talk a lot about immigration, on Chancellor Merkel, and the party itself, as well as share a lot of youtube Videos. CDU users are mostly concerned about the

Fig. 4. Heatmap of user comments on the party pages by topic. The heatmap for hyperactive users aggregates both likes and comments.

Table 3
Hyperactive users per party – Likes & comments.

Party	Likes by Hyperactive Users	Likes Ratio	Comments by Hyperactive Users	Comments Ratio
AfD	555,564	0.35	43,017	0.24
CDU	16,997	0.2	20,929	0.45
CSU	139,493	0.2	18,312	0.22
FDP	20,188	0.16	1,400	0.15
Die Grünen	28,777	0.19	8,946	0.36
Die Linke	24,546	0.14	2,343	0.16
SPD	29,057	0.12	3,926	0.13

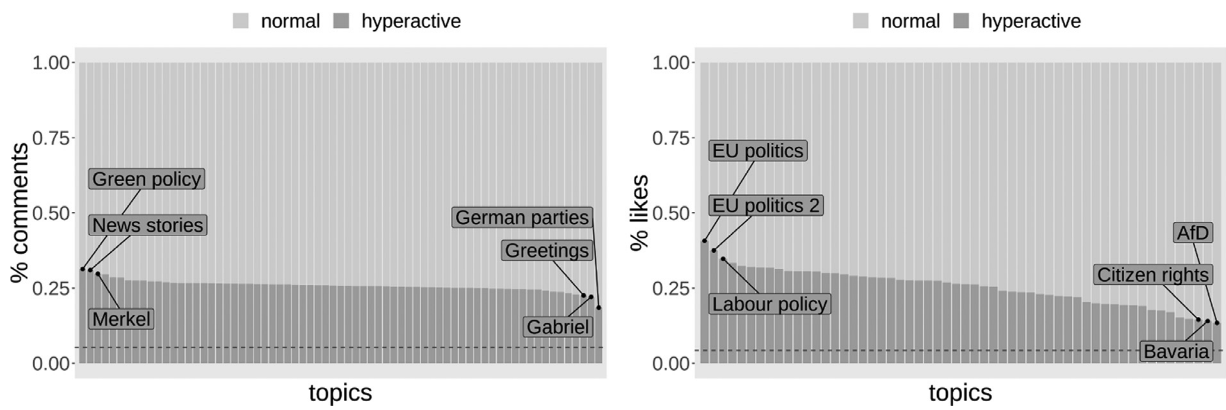

Fig. 5. Proportion of likes and comments generated by normal and hyperactive users. The dotted red line gives the expected proportion of activities for hyperactive users in cases where they were not hyperactive. The plot also illustrates the three most and least interesting topics for hyperactive users.

government and the cooperation with the CSU, topics that are also important for the users on the CSU page. CSU users also discuss a lot about Bavaria, which is the state on which the party is politically based. Users on the FDP page are mostly concerned about citizen rights, while users on the Green Party page about environmental issues. Users on the Left Party page talk about immigration, human rights and welfare policy. Finally, the users on the SPD are focused on party related discussions. Besides illustrating the discussions among pages, the heatmap illustrates how hyperactive users behave. Hyperactive users like and comment more contents related to EU politics and labour policy, and much less on Bavaria, Merkel, citizen rights and on discussions critical against the AfD. It is shown therefore that hyperactive users have their own preferences that do not necessarily concede with the preferences of the rest.

To test if hyperactive users have the same engaging preferences as the rest of the users (H2) we create a stacked chart illustrating the share of hyperactive users' comments and likes for every topic (Fig. 5). It is evident that hyperactive users cover a major part of the comments, and although more active, they comment more or less similarly to the normal users among the various topics. Despite that, the Anderson-Darling tests reject the null hypothesis that hyperactive and normal users' comments come from the same distribution for 54 out of the 69 topics. Practically, this means that the topic density distributions varies between the comments of normal and hyperactive users. This is caused when the comments contain different words in different proportions. Therefore, hyperactive and normal users use different vocabularies when referring to a topic and, consequently, externalize overall different views and sentiment, or focus on different issues in each case.

Besides the fact that hyperactive users have a different behavior on the posts' comments, our analysis shows that they also have different liking preferences. After classifying each party post to the most relevant topic, we count the likes of the posts that belong to each topic. We take into consideration only topics that are based on either political vocabulary or politicians, and ignore topics that contain aggressive speech or sentiments, because the related vocabulary is rarely used by the parties. In contrast to the comments' chart, it is obvious that hyperactive users like specific topics with different intensity than normal users. Even though hyperactive users perform on average 26.4% of the likes, they like much more content related to EU politics and labour policy, while they have less interest on the conservative party AfD, citizens' rights and the region of Bavaria. Therefore, it is clear that hyperactive users influence the like distribution of the public on party posts. Given the statistical analysis of the liking and commenting activities of users, we reject the hypothesis that the distribution of topics in which hyperactive users engage in do not deviate from the distribution of topics in which the rest of the users engage (H2). It must be noted that our analysis gives an overview of the content of posts and comments. It cannot identify sentiment, or specific predispositions of users. For a firm understanding of the issues that are over- or under-represented by hyperactive users an additional extensive analysis is needed, which is beyond the scope of this paper. Our analysis demonstrates that, both on commenting and liking, hyperactive users have a different behavior than the other users.

To further understand hyperactive behavior, we calculate the proportion of comments and likes made by hyperactive users for each post (Fig. 6). Although on average one fourth of the reactions are made by hyperactive users, this ratio is not stable among posts. The proportion of reactions by hyperactive users follows a skewed distribution, denoting that on some posts hyperactive users are almost not active at all, while on some other they are the major content generators. This is especially visible on the distribution of comments, where there are some posts with comments exclusively by hyperactive users, and on a set where hyperactive users are totally absent (around 4% of posts). Most of the posts with no hyperactive users' comments were posts generated on the FDP page. The fact that the above distributions are not normally distributed denotes again that hyperactive users influence asymmetrically the political discourse on the pages. This is an additional finding that reinforces the statement that hyperactive users have an influence on political agenda-setting on OSNs.

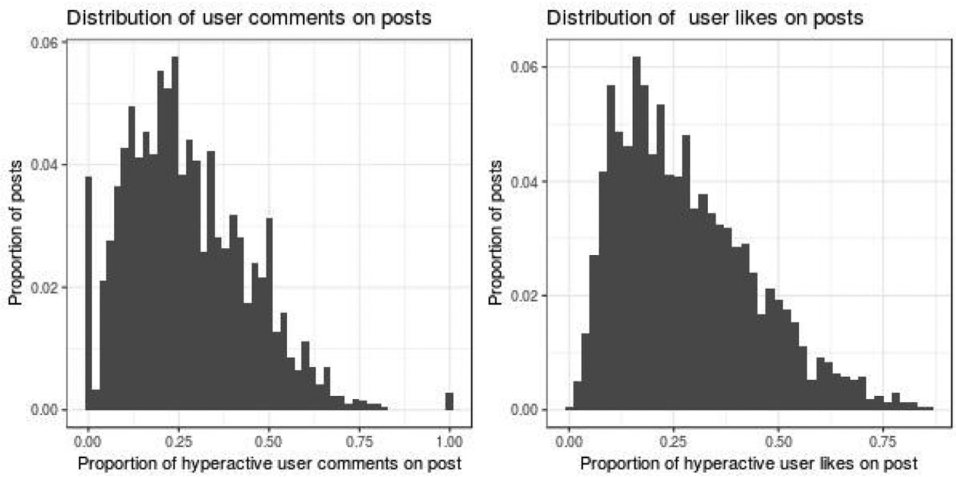

Fig. 6. Distribution of the proportions of comments and likes by hyperactive users on the investigated posts.

Table 4
Accuracy of the various recommender systems for the training and test phase. We report recall@10, NDGC@10 and each model's coverage.

Model	Recall@10 - train	Recall@10 - test	NDGC@10 - train	NDGC@10 - test	coverage - train	coverage - test
CF	0.23	0.2	0.13	0.11	**1**	**1**
HCF	0.24	0.2	0.13	0.13	**1**	0.9
DNN-BWMRB	**0.9**	**0.4**	**0.84**	**0.28**	0.95	0.76

Given the above results, we prove that hyperactive users account for a significant part of the total users' activities (around 25%). They participate in discussions differently from the rest, commenting and liking on different political content than the other users. Moreover, they become opinion leaders, as their comments become more popular than these of the normal users. These findings suggest that hyperactive users have a significant impact on political agenda setting on OSNs. Their preferences and behaviour are different in the discussions taking place, and their opinions are a more valued part of the political discourse.

4.2. Does hyperactive behavior influence recommendation algorithms?

After quantifying the impact of hyperactive users on the political discourse, we illustrate potential effects of hyperactive behaviour on recommendation systems. To do that, we test if the presence of hyperactive users influence recommender systems accuracy on suggesting content that actually corresponds to the political interests of the users (H4). We also test if inserting targeted adversarial attacks on the friendship network (graph poisoning) can result in the distortion of algorithmic recommendations (H5).

Given the extreme skewness of the distribution of user reactions and the high sparsity of the input data (2.9%), we train multiple recommender systems and evaluate their accuracy. We train a baseline collaborative filtering (CF) model that takes only the user friendship network activities as input, a HCF model that has user meta-data and the friendship network activities as input, and a DNN-BWMRB model that has user meta-data, the friendship network activities as inputs, and a cost function accounting for asymmetric user behavior. We split our dataset in train (90%) and test set (10%). To evaluate the models' ability to learn the signal in the dataset and make accurate predictions, we calculate for the training and test set the models' recall@10 and the normalized discounted cumulative gain (NDGC@10). Recall@10 gives the percent of relevant suggestions to the users in the top 10 recommendations. NDGC@10 measures how high in the top 10 recommendations appear the related recommendations (between 0 and 1, with 1 being the best). We also report the coverage of the models, i.e. the percent of classes suggested to the users. Coverage is used to assure that a model does not overfit, predicting only the most popular data classes and not the rest.

Table 4 provides an overview of the results. The baseline CF model yields the worst results, with the HCF providing a slightly better accuracy. Nevertheless, none of the collaborative filtering models is able to learn the true signal in the data, underfitting even at the training phase (CF: Recall@10 - 0.23, HCF: Recall@10 - 0.24). On the contrary, the DNN-BWMRB recommendation system outperforms the other models, being able to make consistent and diverse recommendations (test set: Recall@10 - 0.4, NDGC@10 - 0.28, coverage - 0.76). These values are more than satisfactory, when taking into consideration: (a) the sparsity of the data and the limited amount of observations, (b) the number of labels (69 topics) and the fact that we created our dataset by comprising features from two different user networks.

To further evaluate if the models are able to learn the actual political user interests, we treat the trained systems as generative models and sample from them 69000 recommendations, giving as input the training and the test set. Then, we calculate the proportion of suggestions for each topic and compare it with the actual reactions made by the users. Fig. 7 illustrates how accurately the models learn and replicate the users political interests. As it is visible, the DNN-BWMRB model is able to capture the actual user interests both in the training and test sets, making recommendations that are close to the actual distribution in the data. On the contrary, the HCF only moderately learns the actual distribution of the training set, while failing to generate accurate recommendations for the test set.

The inability of HCF to cope with data sparsity and skewness becomes apparent when sampling recommendations from models with and without hyperactive users (Table 5). While the mean topic proportion is around 1.4%, the M.A.E for the HCF models recommendations is higher (1.7% and 1.9% with and without hyperactive users respectively). This denotes that the models are not able to provide consistent recommendations, proposing significantly different topics to the users for each case. For example,

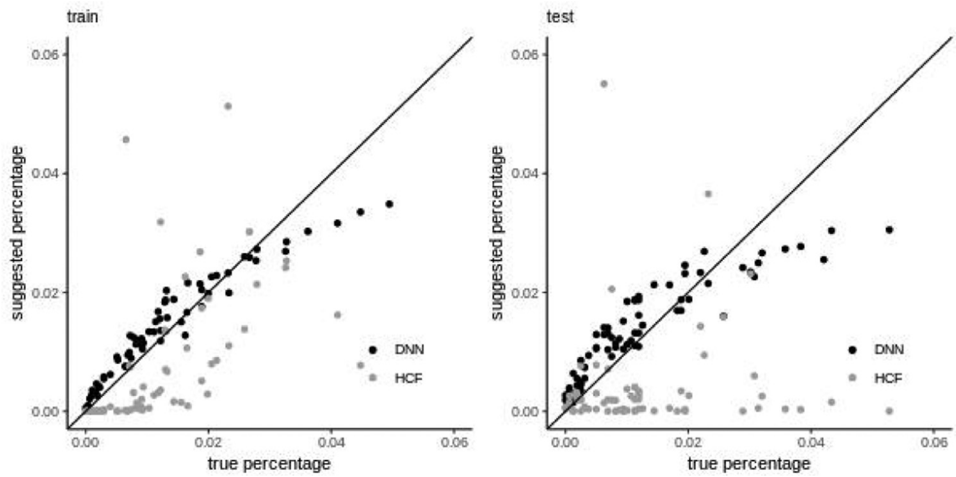

Fig. 7. Accuracy of the two different algorithms trained on the total friendship network containing hyperactive users and taking into consideration n=1403 person related features.

Table 5
Models' accuracy on the test set when taking into consideration or ignoring hyperactive users. The actual mean topic proportion for the users' interests is provided as a reference point.

	w. hyperactive users	w/o hyperactive users
mean topic proportion	0.014	0.014
HCF M.A.E.	0.017	0.019
DNN-BWRB M.A.E.	0.005	0.005

content containing news stories, and issues related to economic, labor, and green poicy would be over-represented by the model trained on all data compared to the model trained only on normal users. The opposite applies for contents related to immigration, citizen rights and youtube videos, which would have been significantly under-represented. To capture which content would have been recommended at each case, we sample posts given the distribution of recommended topics and create wordclouds showing the most prominent tokens for each case (Fig. 8). For the model including hyperactive users, the suggestions would over proportionally contain the token geld (money), together with youtube videos and content about Merkel. It would also suggest content related to the government (Regierung), and immigration (Flüchtling) more often that the model not accounting for hyperactive users. The latter would overproportionally suggest content about justice (Recht), youtube videos, content about Merkel and content containing the token "politik". It would also suggest more often content about the public (Volk) and the police in relation to model including hyperactive users.

Given that, it is clear that such a recommendation algorithm is unable to replicate interactions taking place on OSNs, neither accurately, nor consistently, as including or ignoring hyperactive users yields significantly different recommendations. This fact together with the low accuracy of CF models leads us to confirm H4: Standard recommendation systems fail to suggest content that correspond to the users' political interests, because of the heavy tailed distribution of activities and data sparsity.

On the contrary, the DNN-BWMRB architecture is able not only to provide higher accuracy, but also in a consistent and reliable way. Recommendations made when ignoring or accounting for hyperactive users are similar to each other, with the deviations of predictions from the true distribution of interactions being more or less the same (M.A.E. - 0.005). As the wordclouds in Fig. 8 suggest, both models recommend content related to Merkel, the country, and content from media agencies. They equally promote content related to the police and the euro. Nevertheless, they also illustrate some differences: Token "Politik" is favored by the model with hyperactive users, while token "Partei" is favored by the model without hyperactive users.

Overall, the DNN-BWMRB model is able to capture the signal in the data successfully. Nonetheless, although in a much less degree, the existence of hyperactive users leads the model to provide slightly different suggestions.

As the DNN-BWMRB model is able to capture and replicate the actual user interactions acceptably, we test its predictions when faced with politically malicious attacks (H5). First, we calculate the centrality of each user in the network. Then we insert an adversarial example at the position of the user, which is only engaging to content from topic 10. Next, we calculate the proportion of model suggestions related to topic 10. We repeat the process for different amounts of reactions (10–40) and for different amount of users (1–3). We calculate the bootstrapped mean (50 times, 100 users without replacement) and the maximum value suggested by the model for each setting.

Fig. 9 provides an overview of the results. It is clear that the insertion of adversarial examples influences the suggestions of the recommender systems. The more targeted reactions a malicious user performs, the more relevant suggestions are made by the system. Similarly, the more malicious users we place in the network, the more successful is the distortion of the recommendations to our ends. A simple attack of two users reacting 50 times can lead the recommender system in the network to suggest topic 10 80% of the times. The results of the placed attacks confirm H5, i.e. that graph poisoning can successfully distort recommendation systems suggestions.

By confirming H4 and H5, the study shows that recommendation algorithms are sensitive to hyperactive behaviour. We locate issues regarding the ability of the models to learn from extremely skewed sparse data and make coherent suggestions to the users. We also illustrate how simple it is to manipulate models' inputs in a way that distorts models' recommendations.

5. Discussion

Given that activity asymmetries and recommendation algorithms are a feature of OSNs, it is important to evaluate the consequences for both science and the wider society. Although our analysis was concentrated on Facebook, previous research has proven that hyperactive accounts, either human or automated, have the potential to equally influence political communication on other

(a) HCF with hyperactive users

(b) HCF without hyperactive users

(c) DNN-BWMRB with hyperactive users

(d) DNN-BWMRB without hyperactive users

Fig. 8. Most suggested tokens for different recommender models.

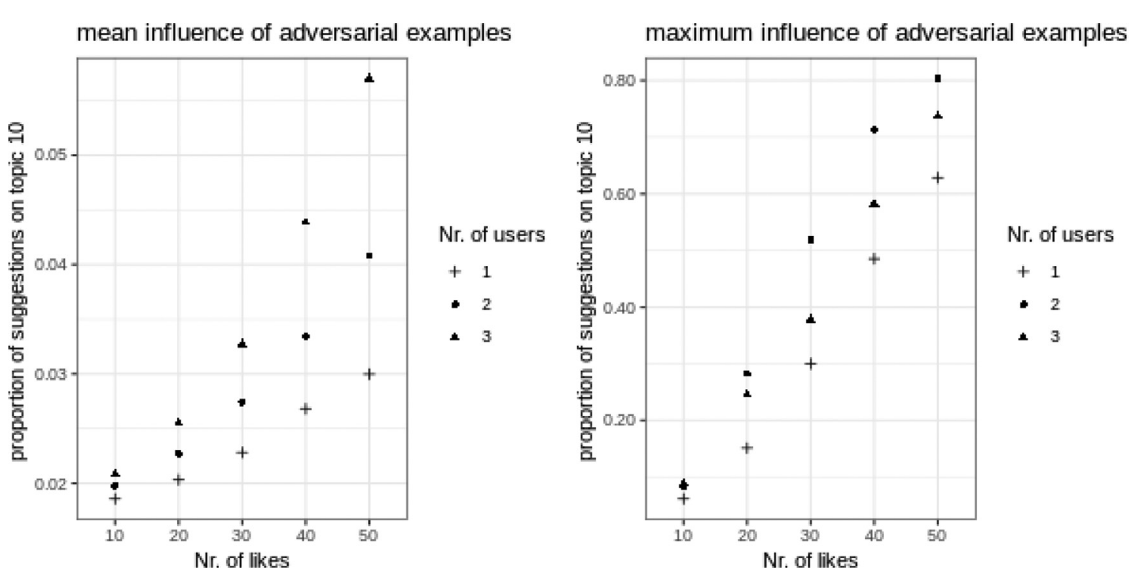

Fig. 9. Mean and maximum proportion of content belonging to Topic 10 after placing adversarial examples.

platforms [70,71]. The specific formation and distribution of political interactions on OSNs raises various questions regarding the role and impact of OSNs on a political level, on an algorithmic level, as well as on the intersection of both.

In the political dimension, the OSN activity asymmetries are transformed into an asymmetry of disseminated political content, as the attitudes and interests of hyperactive users appear disproportionately in the discussions taking place. Until now, research [72,73] has stated that OSNs suffer from a population bias: The people using OSNs are not representative of the actual society. On top of that, a content bias is now added: The content disseminated on OSNs is not even representative of the mean users' attitudes on the platform. We showed that hyperactive users have different attitudes than the rest, and different engaging behavior, altering how the public opinion appears macroscopically. This poses a scientific problem, as it might lead to erroneous research results. Equally important, it poses a political problem, because political discussions and opinion exchange is distorted by the effect of hyperactive accounts. This does not happen because the diffused information in the network is transformed or changed, but because hyperactive users strongly contribute to the type of information diffused. Their attitudes fill the communication space, leading to a bias on the political feedback to politicians, and to a shift on the issues that shape the political agenda. Last but not least, they acquire the status of opinion leaders. Given the above, although OSNs provide a more open environment to express opinions than traditional media, it ends up becoming partly a gathering of political echoes [74] that struggle to be imposed on each other.

In the algorithmic dimension, the extreme skewness of the activity distributions raises specific issues regarding the recommendation algorithms used by OSN platforms. The first problem is related to algorithmic accuracy: skewed data are imbalanced data, and their raw use, either as input features or as output labels, can yield weak classification results. The imbalanced learning problem applies to both standard statistical algorithms, collaborative filtering and neural networks [33,75,76], with algorithms over-estimating the importance of outliers and under-estimating the importance of the rest. This also happens in the case of a poor selection of a cost function [77]. Furthermore, it is proven that statistical models as Markov-chains might fail to capture the signal immanent in highly skewed data, while deep learning methods might face the same issue given power-law distributions of data [34]. We showed that collaborative filtering models come with the same weaknesses. In addition, we showed that the election of a proper cost-function can resolve a large part of the issue.

The second potential problem is that an algorithm might fail, not in the sense that it might be unable to learn from the data, but rather, it might learn the wrong signal. Hyperactive users can be seen as physical adversaries [78] of the mean user attitudes. Algorithms trained in the full data will include the bias, tracking and predicting behavioral associations that correspond to hyperactive users rather than to the population majority. Our study showed that even well trained recommendation systems will have slight deviations in their predictions when accounting for or ignoring hyperactive users. We also illustrated that recommendations of already trained models can easily be distorted by the addition of adversaries of hyperactive behavior. It is not coincidental that the detection of adversaries in machine learning can be done by sample distribution comparison [79], in the same way as we tracked the different preferences of hyperactive users.

Solutions to the aforementioned issues exist, and they were partially shown in the above study and are usually taken into consideration by data scientists, who develop recommendation algorithms. Nevertheless, in the case of political communication, an algorithmic issue automatically becomes a political one. Recommendation systems come with a social influence bias [35,36] i.e., have the power to change users' opinion. Hence, OSNs promoting biased political content might result in the algorithmic manipulation of political communication. This manipulation can be traced back either on the low accuracy of a model, on the attitudes of hyperactive users, or at the structure of the recommender systems themselves.

Social media platforms are not designed to foster political discourses [80], but rather aim at increasing active users, in order to sell advertising and attract funding from venture capitalists [81]. As part of these ends, the structure and impact of recommendation algorithms used distorts human behavior [44], having transformative effects that were not foreseen a priori [82]. For example, it has been stated that changes of the Facebook recommendation algorithm not only resulted in the diffusion of more right-wing content on the platform [20], but also altering in general the way users interact with on the service.

It is evident from the above, that each algorithm mediates and redefines the importance of political interests [83], raising further questions about the opacity of the recommendation systems [84]. In a political context, it becomes important to know as citizens, how, why and with what impact algorithms change political communication. This presupposes awareness of the data processed and, the mathematical method applied, as well as knowledge of what exactly a machine learning cost function optimizes and to what extent recommendation systems alter human behavior. Proposals for algorithmic transparency have already been made [85–87], and wait to be applied in practice.

Until now, most platforms do not report what measures they take to assure the consistency of their recommendations, as well as the exact effect their systems have on human behavior. They do not disclose the inputs and outputs of their models, and provide no information on how recommendations are influenced by malicious organized attacks. Our study tried to illustrate potential effects by quantifying the impact of hyperactive accounts and attacks on simulated case studies. Nevertheless, there is a need for transparency that can illustrate whether the actual recommendation systems used suffer from similar problems.

The above issues need to be extensively analyzed, in order to evaluate and shape the structure of political communication in the digital era. In this paper we laid the foundations for this discussion, by defining, demonstrating and quantifying the effect of hyperactive users on OSNs, through the example of Facebook. We also illustrated and defined the risks of algorithmic manipulation by the OSN recommendation systems. Future research needs to focus on the aforementioned consequences, evaluate the structure of OSNs ethically, politically and normatively as political intermediators, as well as propose and apply solutions to the newly posed problems.

6. Conclusion

Our study showed that political communication on German political Facebook pages is constituted by the behavior of hyperactive users. By describing the users' liking and commenting activities on the pages, we characterized users as hyperactive or normal through outlier detection. We showed that hyperactive users account for a significant part of the total users' activities; they participate in discussions differently from the rest, and they like different content. Moreover, they become opinion leaders, as their comments become more popular than these of the normal users. Taking a set of Facebook pages as an example, we showed that user activities on OSNs might neither be equally nor evenly distributed.

We also studied the influence of hyperactive users on recommendation systems. Standard factorization models are not able to coherently describe the skewed distribution of activities on OSNs. On the other hand, models that balance these asymmetries in their cost function provide results that are closer to the population characteristics. Nevertheless, hyperactive behavior still plays a role on the recommendations generated by the systems. Finally, by adding adversarial examples in the trained models, we illustrated that it is relatively easy to bias recommendations, and potentially distort the political agenda.

Our recommender systems analysis tried to simulate properties of the actual recommender systems used on OSNs. This does not per se mean that the actual systems face the same problems, nor that they function in the exact same way. The aim of the study was to illustrate potential issues related to political communication that might emerge by the application of recommender systems. Given these problems, we want to point out the need for additional transparency on the actual data-intensive recommendation systems used by the platforms, in order to resolve potential algorithmic interferences in the political discourse.

Declaration of Competing Interest

None.

CRediT authorship contribution statement

Orestis Papakyriakopoulos: Writing - review & editing, Writing - original draft, Data curation. **Juan Carlos Medina Serrano:** Data curation, Writing - review & editing. **Simon Hegelich:** Writing - review & editing.

References

[1] J. Gottfried, E. Shearer, Americans' online news use is closing in on tv news use, 2017, http://www.pewresearch.org/fact-tank/2017/09/07/americans-online-news-use-vs-tv-news-use/.
[2] S. Hegelich, M. Shahrezaye, The communication behavior of german MPS on twitter: preaching to the converted and attacking opponents, Eur. Policy Anal. 1 (2) (2015) 155–174.
[3] G.S. Enli, E. Skogerbø, Personalized campaigns in party-centred politics: Twitter and Facebook as arenas for political communication, Inf. Commun. Soc. 16 (5) (2013) 757–774.
[4] V. Arnaboldi, A. Passarella, M. Conti, R. Dunbar, Structure of ego-alter relationships of politicians in twitter, J. Comput. Mediated Commun. 22 (5) (2017) 231–247, doi:10.1111/jcc4.12193.
[5] J.C.M. Serrano, S. Hegelich, M. Shahrezaye, O. Papakyriakopoulos, Social Media Report: The 2017 German Federal Elections, 1 ed., TUM.University Press, 2018.
[6] O. Papakyriakopoulos, S. Hegelich, M. Shahrezaye, J.C.M. Serrano, Social media and microtargeting: Political data processing and the consequences for germany, Big Data Soc. 5 (2) (2018), doi:10.1177/2053951718811844. 2053951718811844
[7] M.E. McCombs, D.L. Shaw, The agenda-setting function of mass media, Public Opin. Q. 36 (2) (1972) 176–187.
[8] E. Bakshy, S. Messing, L.A. Adamic, Exposure to ideologically diverse news and opinion on Facebook, Science 348 (6239) (2015) 1130–1132.
[9] Twitter, 2018, (https://help.twitter.com/en/using-twitter/twitter-trending-faqs). Online; accessed 24 August 2018.
[10] M. Hilbert, J. Vásquez, D. Halpern, S. Valenzuela, E. Arriagada, One step, two step, network step? Complementary perspectives on communication flows in twittered citizen protests, Soc. Sci. Comput. Rev. 35 (4) (2017) 444–461.
[11] S. Choi, The two-step flow of communication in twitter-based public forums, Soc. Sci. Comput. Rev. 33 (6) (2015) 696–711.
[12] C. Mouffe, The Democratic Paradox, Verso, 2000.
[13] N. Blenn, P. Van Mieghem, Are human interactivity times lognormal?, arXiv preprint (2016).
[14] P. Van Mieghem, N. Blenn, C. Doerr, Lognormal distribution in the digg online social network, Eur. Phys. J. B 83 (2) (2011) 251.
[15] K. Lerman, R. Ghosh, Information contagion: An empirical study of the spread of news on digg and twitter social networks, in: Proceedings of the Fourth International AAAI Conference on Web and Social Media, AAAI, 2010, pp. 90–97.
[16] F. Benevenuto, T. Rodrigues, M. Cha, V. Almeida, Characterizing user behavior in online social networks, in: Proceedings of the 9th ACM SIGCOMM Conference on Internet Measurement, ACM, 2009, pp. 49–62.
[17] A. Schaap, Agonism in divided societies, Philos. Soc. Crit. 32 (2) (2006) 255–277.
[18] N. Shi, M.K. Lee, C.M. Cheung, H. Chen, The continuance of online social networks: how to keep people using Facebook? in: Proceedings of the Fourty-Third Hawaii International Conference on System Sciences, IEEE, 2010, pp. 1–10.
[19] C.R. Sunstein, # Republic: Divided Democracy in the Age of Social Media, Princeton University Press, 2018.
[20] N. Analytics, The 2019 guide to Facebook publishing, http://go.newswhip.com/2019_03-FacebookPublishing_LP.html.
[21] T. Bucher, Want to be on the top? Algorithmic power and the threat of invisibility on Facebook, New Media Soc. 14 (7) (2012) 1164–1180.
[22] E.D. Hersh, Hacking the Electorate: How Campaigns Perceive Voters, Cambridge University Press, 2015.
[23] C. Boldrini, M. Toprak, M. Conti, A. Passarella, Twitter and the press: an ego–centred analysis, in: Proceedings of the The Web Conference, International World Wide Web Conferences Steering Committee, 2018, pp. 1471–1478.
[24] D.M. Romero, W. Galuba, S. Asur, B.A. Huberman, Influence and passivity in social media, in: Proceedings of the Joint European Conference on Machine Learning and Knowledge Discovery in Databases, Springer, 2011, pp. 18–33.
[25] B.E. Weeks, A. Ardèvol-Abreu, H. Gil de Zúñiga, Online influence? Social media use, opinion leadership, and political persuasion, Int. J. Public Opin. Res. 29 (2) (2017) 214–239.
[26] V. Arnaboldi, M. Conti, A. Passarella, R.I. Dunbar, Online social networks and information diffusion: the role of ego networks, Online Soc. Netw. Media 1 (2017) 44–55.
[27] S. Priya, R. Sequeira, J. Chandra, S.K. Dandapat, Where should one get news updates: Twitter or Reddit, Online Soc. Netw. Media 9 (2019) 17–29.
[28] S. Hegelich, D. Janetzko, Are social bots on twitter political actors? Empirical evidence from a ukrainian social Botnet, in: Proceedings of the Tenth International AAAI Conference on Web and Social Media, 2016, pp. 1–4.
[29] J. Weedon, W. Nuland, A. Stamos, Information operations and Facebook, Retrieved from Facebook: https://fbnewsroomus.files.wordpress.com/2017/04/facebook-and-information-operations-v1.pdf (2017).
[30] P. Covington, J. Adams, E. Sargin, Deep neural networks for Youtube recommendations, in: Proceedings of the 10th ACM Conference on Recommender Systems, ACM, 2016, pp. 191–198.
[31] The Facebook algorithm explained and how to work it, https://www.brandwatch.com/blog/the-facebook-algorithm-explained/.
[32] Q. Yuan, S. Zhao, L. Chen, Y. Liu, S. Ding, X. Zhang, W. Zheng, Augmenting collaborative recommender by fusing explicit social relationships, in: Proceedings of the Workshop on Recommender Systems and the Social Web, RECSYS, 2009, pp. 1–8.
[33] H. He, E.A. Garcia, Learning from imbalanced data, IEEE Trans. Knowl. Data Eng. 9 (2008) 1263–1284.
[34] H.W. Lin, M. Tegmark, Criticality in Formal Languages and Statistical Physics, Entropy 19 (2017) 299, doi:10.3390/e19070299.
[35] S. Krishnan, J. Patel, M.J. Franklin, K. Goldberg, A methodology for learning, analyzing, and mitigating social influence bias in recommender systems, in: Proceedings of the 8th ACM Conference on Recommender systems, ACM, 2014, pp. 137–144.
[36] D. Cosley, S.K. Lam, I. Albert, J.A. Konstan, J. Riedl, Is seeing believing?: How recommender system interfaces affect users' opinions, in: Proceedings of the SIGCHI Conference on Human Factors in Computing Systems, ACM, 2003, pp. 585–592.
[37] X. Chen, C. Liu, B. Li, K. Lu, D. Song, Targeted Backdoor Attacks on Deep Learning Systems Using Data Poisoning, http://arxiv.org/abs/1712.05526 [Cs] (2017). (accessed December 19, 2019).
[38] B. Li, Y. Wang, A. Singh, Y. Vorobeychik, Data poisoning attacks on factorization-based collaborative filtering, in: Proceedings of the Advances in Neural Information Processing Systems, 2016, pp. 1885–1893.
[39] A. Bojchevski, S. Günnemann, Adversarial attacks on node embeddings via graph poisoning, in: Proceedings of the International Conference on Machine Learning, 2019, pp. 695–704.
[40] D. Zügner, S. Günnemann, Adversarial Attacks on Graph Neural Networks via Meta Learning, http://arxiv.org/abs/1902.08412 [Cs, Stat] (2019). (accessed December 19, 2019).
[41] Facebook, Using the graph API – documentation, https://developers.facebook.com/docs/graph-api/using-graph-api/.
[42] O. Varol, E. Ferrara, C.A. Davis, F. Menczer, A. Flammini, Online human-bot interactions: detection, estimation, and characterization, in: Proceedings of the Eleventh International AAAI Conference on Web and Social Media, 2017, pp. 1–7.
[43] J.C.M. Serrano, M. Shahrezaye, O. Papakyriakopoulos, S. Hegelich, The rise of Germany's AFD: a social media analysis, in: Proceedings of the 10th International Conference on Social Media and Society, in: SMSociety '19, ACM, New York, NY, USA, 2019, pp. 214–223, doi:10.1145/3328529.3328562.
[44] D. Ruths, J. Pfeffer, Social media for large studies of behavior, Science 346 (6213) (2014) 1063–1064.
[45] T.B. Arnold, J.W. Emerson, Nonparametric goodness-of-fit tests for discrete null distributions., R J. 3 (2) (2011).
[46] Q.H. Vuong, Likelihood ratio tests for model selection and non-nested hypotheses, Econom. J. Econom. Soc. 57 (2) (1989) 307–333.
[47] A. Clauset, C.R. Shalizi, M.E. Newman, Power-law distributions in empirical data, SIAM review 51 (4) (2009) 661–703.
[48] V. Barnett, T. Lewis, Outliers in Statistical Data, Wiley, 1974.
[49] R. Johnson, D. Wichern, Applied Multivariate Statistical Analysis., Pretice Hall, 1998.
[50] I. Ben-Gal, Outlier detection, in: Data Mining and Knowledge Discovery Handbook, Springer, 2005, pp. 131–146.
[51] M. Hubert, S. Van der Veeken, Outlier detection for skewed data, J. Chemom. J. Chemom. Soc. 22 (3–4) (2008) 235–246.
[52] M. Hubert, E. Vandervieren, An adjusted boxplot for skewed distributions, Comput. Stat. Data Anal. 52 (12) (2008) 5186–5201.

[53] G. Brys, M. Hubert, A. Struyf, A robust measure of skewness, J. Comput. Gr. Stat. 13 (4) (2004) 996–1017.
[54] F. Martin, M. Johnson, More efficient topic modelling through a noun only approach, in: Proceedings of the Australasian Language Technology Association Workshop 2015, 2015, pp. 111–115.
[55] M. Honnibal, M. Johnson, An improved non-monotonic transition system for dependency parsing, in: Proceedings of the Conference on Empirical Methods in Natural Language Processing, 2015, pp. 1373–1378.
[56] S. Brants, S. Dipper, P. Eisenberg, S. Hansen-Schirra, E. König, W. Lezius, C. Rohrer, G. Smith, H. Uszkoreit, Tiger: linguistic interpretation of a German corpus, Res. Lang. Comput. 2 (4) (2004) 597–620.
[57] J. Nothman, N. Ringland, W. Radford, T. Murphy, J.R. Curran, Learning multilingual named entity recognition from wikipedia, Artif. Intell. 194 (2013) 151–175.
[58] M. Yurochkin, A. Guha, X. Nguyen, Conic scan-and-cover algorithms for nonparametric topic modeling, in: Proceedings of the Advances in Neural Information Processing Systems, 2017, pp. 3878–3887.
[59] D.M. Blei, A.Y. Ng, M.I. Jordan, Latent Dirichlet allocation, J. Mach. Learn. Res. 3 (2003) 993–1022.
[60] D.D. Lee, H.S. Seung, Learning the parts of objects by non-negative matrix factorization, Nature 401 (6755) (1999) 788.
[61] F.W. Scholz, M.A. Stephens, K-sample Anderson–darling tests, J. Am. Stat. Assoc. 82 (399) (1987) 918–924.
[62] J. Leskovec, J.J. Mcauley, Learning to discover social circles in ego networks, in: Proceedings of the Advances in Neural Information Processing Systems, 2012, pp. 539–547.
[63] M. McPherson, L. Smith-Lovin, J.M. Cook, Birds of a feather: homophily in social networks, Annu. Rev. Sociol. 27 (1) (2001) 415–444.
[64] J. Kirk, Tensorrec: a tensorflow recommendation algorithm and framework in python, 2019, https://github.com/jfkirk/tensorrec [Online; accessed 04.05.2019].
[65] J. Constine, How facebook news feed works techcrunch, 2016, https://techcrunch.com/2016/09/06/ultimate-guide-to-the-news-feed/
[66] Twitter trends Faqs, https://help.twitter.com/en/using-twitter/twitter-trending-faqs.
[67] K. Liu, P. Natarajan, WMRB: Learning to Rank in a Scalable Batch Training Approach, http://arxiv.org/abs/1711.04015 [Cs, Stat] (2017). (accessed December 19, 2019).
[68] K. Sun, Explanation of log-normal distributions and power-law distributions in biology and social science, Technical Report, Department of Physics, University of Illinois at Urbana-Champaign, 2004.
[69] H. Kuninaka, M. Matsushita, Statistical properties of complex systems-lognormal and related distributions, in: Proceedings of the AIP Conference Proceedings, 1468, AIP, 2012, pp. 241–251.
[70] A. Thieltges, O. Papakyriakopoulos, J.C.M. Serrano, S. Hegelich, Effects of Social Bots in the Iran-Debate on Twitter, http://arxiv.org/abs/1805.10105 [Cs] (2018). (accessed December 19, 2019).
[71] C. Shao, G.L. Ciampaglia, O. Varol, K.-C. Yang, A. Flammini, F. Menczer, The spread of low-credibility content by social bots, Nat. Commun. 9 (1) (2018) 4787.
[72] T. Correa, A.W. Hinsley, H.G. De Zuniga, Who interacts on the web?: the intersection of users personality and social media use, Comput. Hum. Behav. 26 (2) (2010) 247–253.
[73] M. Duggan, J. Brenner, The demographics of social media users - 2012, Technical Report, Pew Research Center's Internet & American Life Project Washington, DC, 2013.
[74] Y.-R. Lin, J.P. Bagrow, D. Lazer, More voices than ever? Quantifying media bias in networks, in: Proceedings of the Fifth International AAAI Conference on Web and Social Media, AAAI, 2011, pp. 1–7.
[75] Z.-H. Zhou, X.-Y. Liu, Training cost-sensitive neural networks with methods addressing the class imbalance problem, IEEE Trans. Knowl. Data Eng. 18 (1) (2006) 63–77.
[76] R. Pan, Y. Zhou, B. Cao, N.N. Liu, R. Lukose, M. Scholz, Q. Yang, One-class collaborative filtering, in: Proceedings of the IEEE Eighth International Conference on Data Mining, IEEE, 2008, pp. 502–511.
[77] C.-C. Lee, P.-C. Chung, J.-R. Tsai, C.-I. Chang, Robust radial basis function neural networks, IEEE Trans. Syst. Man Cybern. Part B (Cybernetics) 29 (6) (1999) 674–685.
[78] A. Kurakin, I. Goodfellow, S. Bengio, Adversarial examples in the physical world, http://arxiv.org/abs/1607.02533 [Cs, Stat] (2017). (accessed December 19, 2019).
[79] K. Grosse, P. Manoharan, N. Papernot, M. Backes, P. McDaniel, On the (Statistical) Detection of Adversarial Examples, http://arxiv.org/abs/1702.06280 [Cs, Stat] (2017). (accessed December 19, 2019).
[80] S. Engelmann, J. Grossklags, O. Papakyriakopoulos, A democracy called Facebook? Participation as a privacy strategy on social media, in: Proceedings of the Annual Privacy Forum, Springer, 2018, pp. 91–108.
[81] M. Falch, A. Henten, R. Tadayoni, I. Windekilde, Business models in social networking, in: Proceedings of the CMI International Conference on Social Networking and Communities, 2009, pp. 1–7.
[82] B.D. Mittelstadt, P. Allo, M. Taddeo, S. Wachter, L. Floridi, The ethics of algorithms: Mapping the debate, Big Data Soc. 3 (2) (2016) 2053951716679679.
[83] H. Nissenbaum, From preemption to circumvention: if technology regulates, why do we need regulation (and vice versa), Berk. Technol. Law J. 26 (2011) 1367.
[84] J. Burrell, How the machine thinks: understanding opacity in machine learning algorithms, Big Data Soc. 3 (1) (2016) 2053951715622512.
[85] C. Sandvig, K. Hamilton, K. Karahalios, C. Langbort, Auditing algorithms: research methods for detecting discrimination on internet platforms, in: Data and Discrimination: Converting Critical Concerns into Productive Inquiry, 2014, pp. 1–23.
[86] N. Diakopoulos, Algorithmic-accountability: the investigation of Black Boxes, Technical Report, Tow Center for Digital Journalism, 2014.
[87] F. Pasquale, The Black Box Society: The Secret Algorithms that Control Money and Information, Harvard University Press, 2015.